Sequences & Series

ANSWER KEY

Section 1: Advanced Arithmetic & Geometric Progressions

1 The sum of three consecutive terms of an arithmetic progression is 27, and their product is 504. What is the largest of the three terms?

a. 14 ✓

b. 16

c. 18

d. 21

Let terms be $a-d, a, a+d$. Sum $= 3a = 27 \Rightarrow a = 9$. Product $= (9-d)(9)(9+d) = 9(81-d^2) = 504 \Rightarrow 81 - d^2 = 56 \Rightarrow d^2 = 25 \Rightarrow d = 5$. Largest term $= 9 + 5 = 14$.

2 If x, y, z are in geometric progression, which equation must be true?

a. $y^2 = xz$ ✓

b. $2y = x + z$

c. $y = \frac{x+z}{2}$

d. $y^3 = xz$

Common ratio: $\frac{y}{x} = \frac{z}{y} \Rightarrow y^2 = xz$.

3 What is the sum of the series $1 + \frac{2}{3} + \frac{4}{9} + \frac{8}{27} + \ldots$?

a. 3 ✓

b. $\frac{3}{2}$

c. 2

d. $\frac{5}{2}$

$a = 1, r = \frac{2}{3}$. Sum $= \frac{1}{1-2/3} = \frac{1}{1/3} = 3$.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

Section 2: Summation Notation & Formulas

4 Evaluate $\sum_{k=1}^{15} (4k - 1)$.

a. 465 ✓

b. 480

c. 495

d. 510

Sum $= 4\sum_{k=1}^{15} k - 15(1) = 4(\frac{15 \times 16}{2}) - 15 = 4(120) - 15 = 480 - 15 = 465$.

5 Evaluate $\sum_{k=1}^{8} k^3$ using the formula $[\frac{n(n+1)}{2}]^2$.

a. $1,296$ ✓

b. $1,024$

c. $1,521$

d. $1,728$

$[\frac{8 \times 9}{2}]^2 = [36]^2 = 1,296$.

6 Evaluate $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$.

a. 1 ✓

b. $\frac{1}{2}$

c. 2

d. Diverges

Telescoping series: $\sum(\frac{1}{n} - \frac{1}{n+1}) = 1 - 0 = 1$.

Section 3: Infinite Geometric Series & Repeating Decimals

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

7 A ball is dropped from a height of 16 meters. Each time it hits the ground, it bounces to $\frac{3}{4}$ of its previous height. What is the total vertical distance traveled before coming to rest?

a. 112 **meters** ✓

b. 96 meters

c. 128 meters

d. 64 meters

Distance $= 16 + 2\sum_{n=1}^{\infty} 16(3/4)^n = 16 + 2[\frac{12}{1-3/4}] = 16 + 2(48) = 16 + 96 = 112$ meters.

8 Express $0.\overline{36}$ as a reduced fraction.

a. $\frac{4}{11}$ ✓

b. $\frac{36}{100}$

c. $\frac{1}{3}$

d. $\frac{12}{33}$

Sum of geometric series: $\frac{36/100}{1-1/100} = \frac{36}{99} = \frac{4}{11}$.

9 For what range of x does $\sum_{n=0}^{\infty} (\frac{x}{3})^n$ converge?

a. $(-3, 3)$ ✓

b. $[-3, 3]$

c. $(-1, 1)$

d. $(-9, 9)$

Convergence requires $|r| < 1$: $|\frac{x}{3}| < 1 \Rightarrow -3 < x < 3$.

Section 4: Mathematical Induction

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

10 In proving $1 + 2 + \ldots + n = \frac{n(n+1)}{2}$, what is the inductive step?

a. **Add $(k+1)$ to both sides of the hypothesis for $n = k$** ✓

b. Multiply both sides by 2

c. Subtract k from n

d. Differentiate with respect to n

The inductive step assumes the identity for $n = k$ and adds the next term $(k+1)$ to prove the formula for $k+1$.

11 What is the base case when proving a theorem for all positive integers n?

a. **Verify the statement for $n = 1$** ✓

b. Verify for $n = 0$

c. Assume for $n = k$

d. Prove for all even integers

The base case establishes the initial truth of the proposition, typically at $n = 1$.

12 If $P(k) \Rightarrow P(k+1)$ is true, under what condition is $P(n)$ true for all $n \geq 1$?

a. $P(1)$ **is true** ✓

b. $P(0)$ is true

c. $P(2)$ is false

d. $P(k)$ is true for all k simultaneously

By the Principle of Mathematical Induction, if the base case $P(1)$ holds and $P(k) \Rightarrow P(k+1)$, the proposition holds for all $n \geq 1$.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647